

## [2.6] Specification Gaming & Goal Misgeneralisation

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ILIAD Intensive

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Even with a *perfectly specified* reward, it is possible for the agent to still internalise the wrong goal.

# The pottery shop

A grid world: a robot shares the floor with fragile **urns** and piles of **shards**, plus a **bin**. We want it to tidy shards into the bin without smashing urns.

- Actions: move around, PICKUP, PUTDOWN.
- Walking into an urn breaks it into shards.
- Shards are disposed of by picking them up, and dropping them in the bin.
- Play this environment in your browser:  
<https://iliad-intensive.org/uploads/policy-gradients-misgeneralization/play.html>



# Reward functions as the interface

## The reward hypothesis

“That all of what we mean by goals and purposes can be well thought of as maximization of the expected value of the cumulative sum of a received scalar signal (reward).” – Richard Sutton

Assuming this hypothesis, it should be possible (at least in principle) to encourage any sort of behaviour we want by finding just the right reward function.

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We want to train the agent to pick shards up and carry them to the bin. A first attempt, reward1:

- 1 +1 for picking up shards,
- 2 +1 for dropping them in the bin.

What behaviour does this reinforce in practice?



# Specification gaming: the agent farms reward

Trained on reward1, the agent discovers a loophole:

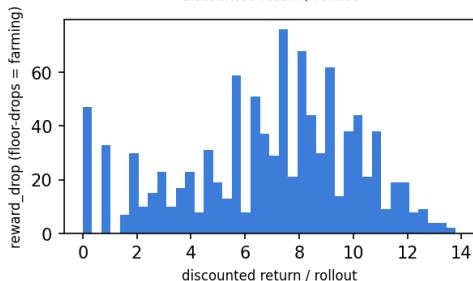
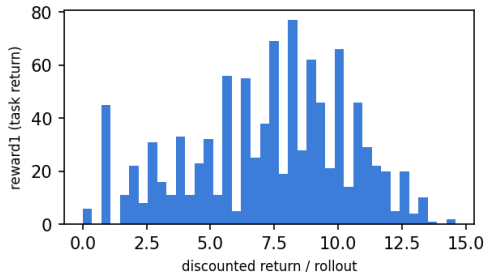
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We probe behaviour with extra reward functions: reward\_drop (+1 per floor-drop). Its return tracks reward1 almost exactly.



## Fix 1: potential shaping (the un-hackable hint)

We still want to *hint* that holding shards is good, without creating a loophole.  
Removing the reward from picking up shards makes the task very hard to learn.

**Potential shaping** (Ng, Harada & Russell, 1999) [NHR99]: pick a bounded  $\Phi : \mathcal{S} \rightarrow \mathbb{R}$  and define a new reward function

$$R^\Phi(s, a, s') = R(s, a, s') + \gamma \Phi(s') - \Phi(s).$$

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Here  $\Phi(s) = 0.5$  while holding shards. A pickup advances  $+\gamma\Phi$ ; a floor-drop gives it back. A **pickup-then-drop cycle nets exactly zero**: the loophole is closed, while also encouraging the agent to try picking up shards in the first place.

# Why shaping preserves the optimum

Write the return from time  $t$  with  $r_k := R(s_{k-1}, a_{k-1}, s_k)$ :

$$G_t = r_{t+1} + \gamma r_{t+2} + \gamma^2 r_{t+3} + \cdots = \sum_{k=t+1}^{\infty} \gamma^{k-(t+1)} r_k.$$

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Hence  $G_t^\Phi = G_t - \Phi(s_t)$ . At  $t = 0$ ,  $G_0^\Phi = G_0 - \Phi(s_0)$  differs from  $G_0$  by a constant depending *only on the start state*, not on the policy: so  $\arg \max_{\pi} G_0^\Phi = \arg \max_{\pi} G_0$  and the optimum is unchanged.

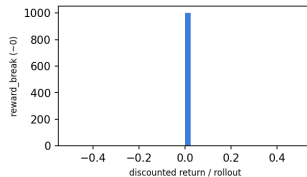
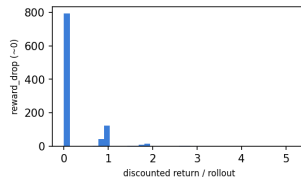
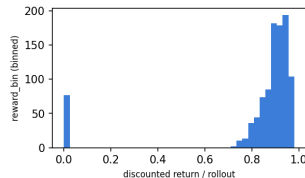
# Specification gaming fixed?

Shaping stops farming, but nothing yet discourages breaking urns to manufacture shards.

- Add a  $-2$  penalty for each urn smashed.
- Binning the resulting shard recovers  $\approx +1$  (discounted), against a  $-2$  penalty: so breaking-then-cleaning is never worth it.

Retrained on reward2, the probes confirm the fix:

- reward\_drop (farming)  $\approx 0$ ,
- reward\_break (smashing)  $\approx 0$ ,
- reward\_bin  $> 0$ : it actually cleans.



# Motivating the potential function

- Why bother with the potential function at all?  $+1$  reward for putting shards in the bin, and  $-2$  for smashing urns, also is aligned with the intended goal.

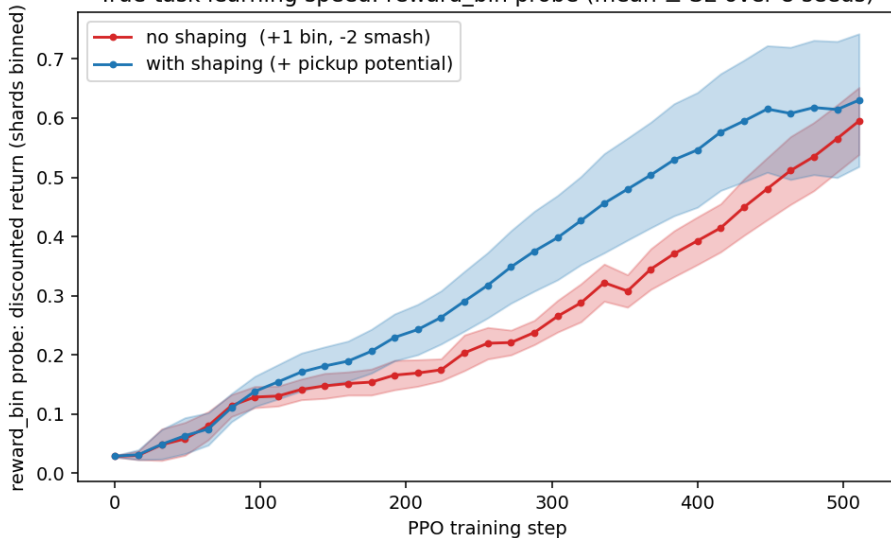
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- But the agent still has to learn from experience: the policy does not teleport to whatever maximises the sum of rewards.
- Leaving the potential function out robs the agent of useful feedback (that picking up and walking around with shards is instrumentally useful for putting them in the bin).

True-task learning speed: reward\_bin probe (mean  $\pm$  SE over 8 seeds)



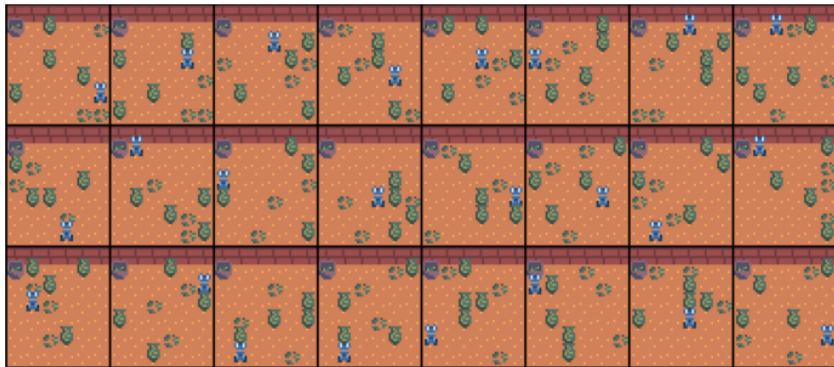
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So we train on **procedurally generated** layouts: random robot and item positions every episode. As a toy example of insufficient training environment diversity, suppose that the bin is always in the top-left corner.





# Goal misgeneralisation: defining it

**Langosco et al. (2022)** [LKS<sup>+</sup>22]: an agent trained to maximise  $R$  is deployed under distribution shift. *Goal misgeneralisation* occurs if it now gets **low reward** (according to  $R$ ) because it keeps acting as if it was trained to optimise a *different*  $R' \neq R$ .

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- $R$  = **intended objective** (carry shards to the bin),
- $R'$  = **behavioural objective** (what it actually learned).

During training, “go to the bin” and “go to the top-left corner” were **indistinguishable**. The simpler one can win, and the model has internalised the wrong goal.

# Out of distribution: move the bin

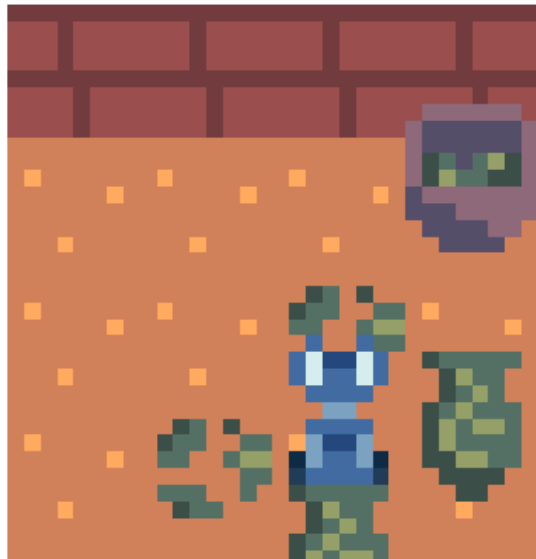
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## Out of distribution: move the bin

At test time, put the bin somewhere else (e.g. the top-right corner). The agent still capably picks up shards and carries them **to the top-left corner**, where it drops them on the floor, ignoring the actual bin. Defining a proxy reward function  $R'$  that rewards dropping shards at the top left corner, we can probe the agent's behaviour on the shifted layout.



# Goal misgeneralisation, measured

On shifted layouts, the agent:

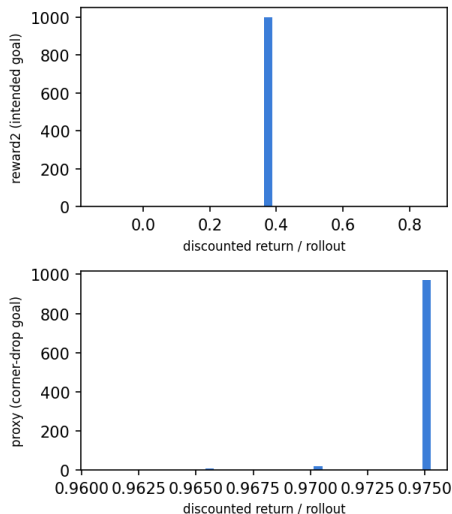
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It is still competently optimising a goal, just not what we trained for. There is no black-box way to tell apart an agent that learned  $R$  from one that learned  $R'$  as the behaviour is the same on in-distribution layouts.



# Fixing it: broaden the training distribution

- Randomise the bin position too.
- Now “corner” no longer correlates with “bin”, so the agent is forced to learn the intended goal  $R$ .

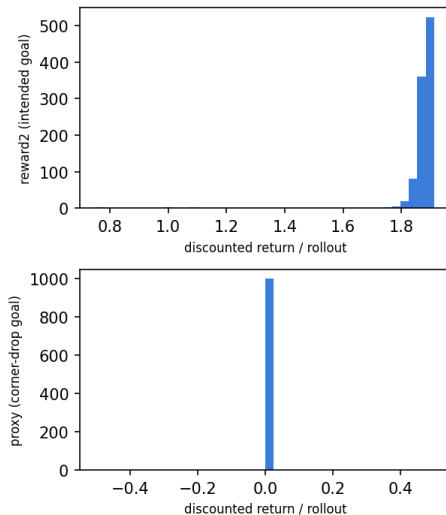


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- Result: The training distribution now matches the test distribution, so the agent is no longer misgeneralising.
- **Caveat:** In practice you often cannot enumerate the shift in advance, so this fix is not always available. Mitigating goal misgeneralisation without it is an open problem.



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- Broadening the training distribution helps, when you can; and “reward-maximising” is necessary but not sufficient for “learned”.

# References I

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